CS6700 — Reinforcement Learning

Tier	B
Phase (order)	7
Do this first	CS4011, probability
Estimated effort	12 weeks
Unlocks	DA7400, CS7011

Roadmap

1. Bandits; MDPs; dynamic programming.
2. Monte Carlo, TD, Q-learning, SARSA.
3. Function approximation, DQN, policy gradients, actor-critic.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Reinforcement Learning (64 videos). NPTEL *Reinforcement Learning* (Balaraman Ravindran, IIT Madras): this is CS6700's teacher.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- DeepMind x UCL | Introduction to Reinforcement Learning 2015 (10 videos). DeepMind x UCL *Introduction to RL 2015* (David Silver, official Google DeepMind): the classic. Pair with Sutton & Barto.
- Stanford CS234 I Reinforcement Learning I Spring 2024 I Emma Brunskill (16 videos). Stanford CS234 Spring 2024 (Emma Brunskill, official): modern, includes offline RL and RLHF.
- Special topics in ML (Reinforcement Learning) (44 videos). IIT Madras BS *Special topics in ML (Reinforcement Learning)*.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–6	Probability and linear algebra tutorials, RL framework.
2-3	7–18	Bandits: UCB, PAC bounds, median elimination, Thompson sampling, policy search, contextual bandits. Implement epsilon-greedy, UCB and Thompson on a 10-armed testbed (Sutton & Barto ch. 2).
4	19–27	MDPs, Bellman equations, contraction / Banach fixed point proofs.

Week	Lectures	What to do
5	28–30	Value iteration, policy iteration, DP. Solve FrozenLake with both.
6	31–38	Monte Carlo, off-policy MC, UCT, TD(0), SARSA, Q-learning. Implement Q-learning on CliffWalking (Gymnasium).
7	39–42	Eligibility traces, TD(lambda).
8	43–49	Function approximation, LSTD, LSPI, fitted Q, DQN. Train a DQN on CartPole.
9	50–53	Policy gradients, REINFORCE, actor-critic. Implement REINFORCE and A2C.
10	54–64	Hierarchical RL (options, MAXQ), POMDPs. Then PPO from Spinning Up (OpenAI).

▼ **All lectures in *Reinforcement Learning* (64 videos)**

1. Tutorial 1 - Probability Basics 1
2. Tutorial 1 - Probability Basics 2
3. Tutorial 2 - Linear Algebra 1
4. Tutorial 2 - Linear Algebra 2
5. Introduction to RL
6. RL Framework and Applications
7. Introduction to Immediate RL
8. Bandit Optimalities
9. Value Function Based Methods
10. UCB 1
11. Concentration Bounds
12. UCB 1 Theorem
13. PAC Bounds
14. Median Elimination
15. Thompson Sampling
16. Policy Search
17. REINFORCE
18. Contextual Bandits
19. Full RL Introduction
20. Returns, Value functions and MDPs
21. MDP Modelling
22. Bellman Equation
23. Bellman Optimality Equation
24. Lec 20 - Cauchy Sequence and Green's Equation
25. Banach Fixed Point Theorem
26. Convergence Proof
27. Lpi Convergence
28. Value Iteration
29. Policy Iteration

30. Dynamic Programming
31. Monte Carlo
32. Control in Monte Carlo
33. Off Policy MC
34. UCT
35. TD(0)
36. TD(0) Control
37. Lec 33 - Q-Learning
38. Lec 34 - Afterstate
39. Eligibility Traces
40. Backward View of Eligibility Traces
41. Eligibility Trace Control
42. Thompson Sampling
43. Function Approximation
44. Linear Parameterization
45. State Aggregation Methods
46. Function Approximation and Eligibility Traces
47. LSTD and LSTDQ
48. LSPI and Fitted Q
49. DQN and Fitted Q Iteration
50. Policy Gradient Approach
51. Actor Critic and REINFORCE
52. REINFORCE (cont'd)
53. Policy Gradient with Function Approximation
54. Hierarchical Reinforcement Learning
55. Types of Optimality
56. Semi Markov Decision Processes
57. Options
58. Learning with Options
59. Hierarchical Abstract Machines
60. MAXQ
61. MAXQ Value Function Decomposition
62. Option Discovery
63. POMDP Introduction
64. Solving POMDP

Books, notes and websites

- ✅ NPTEL *Reinforcement Learning* (Balaraman Ravindran, IIT Madras) — onlinecourses.nptel.ac.in/noc20_cs74/preview
- Book: Sutton & Barto — incompleteideas.net/book (free; did not load from my check, search the title if it is down)

- David Silver's UCL lectures (YouTube); Hugging Face Deep RL course

Practice / proof of understanding

Solve FrozenLake with value iteration, CartPole with DQN, and LunarLander with PPO using Gymnasium.

What your IITM course sheet says

From the PDF: CS6700

Official description: The Reinforcement Learning problem : evaluative feedback, non-associative learning, Rewards and returns, Markov Decision Processes, Value functions, optimality and approximation. Dynamic programming : value iteration, policy iteration, asynchronous DP, generalized policy iteration. Monta-Carlo methods : policy evaluation, roll outs, on policy and off policy learning, importance sampling. Temporal Difference learning : TD prediction, Optimality of TD(0), SARSA, Q-learning, R-learning, Games and after states. Eligibility traces : n-step TD prediction, TD (lambda), forward and backward views, Q (lambda), SARSA (lambda), replacing traces and accumulating traces. Function Approximation : Value prediction, gradient descent methods, linear function approximation, ANN based function approximation, lazy learning, instability issues Policy Gradient methods : non-associative learning

Official textbooks: NIL